

Lec 9 Almgren-Chriss, Trading with Impact, and the TWAP Strategy

Large liquidation problem:

$\mathcal{N}$ = # of shares

to be liquidated in $[0, T]$.

Inventory process Q_t ,

$$dQ_t = -v_t dt$$

where v_t is the control,

with $Q_0 = \mathcal{N}$ and $Q_T = 0$.

The issue is price impact,

Permanent Impact $g(v)$:

shifts the mid-price,

$$dS_t = -g(v_t)dt + \sigma dW_t,$$

a reduced form price impact.

Temporary Impact $f(v)$:

instantaneous order book

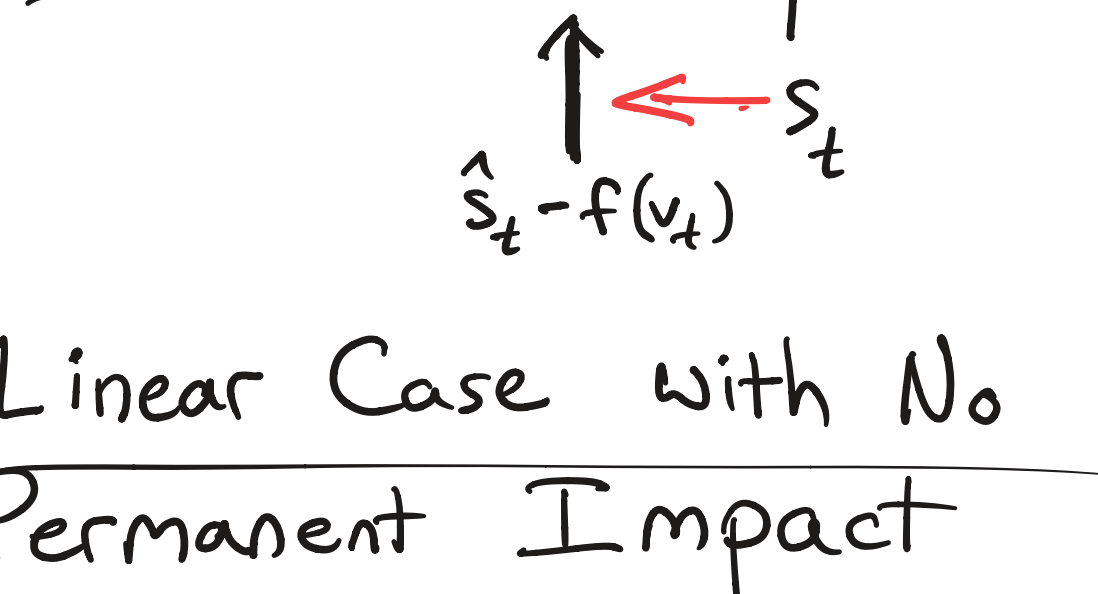
slippage,

$$\hat{S}_t = S_t - f(v_t).$$

Convenient specification

is $f(v) = kv$

$$\Rightarrow \hat{S}_t = S_t - kv_t.$$



Linear Case with No Permanent Impact

$$f(v) = kv \quad g(v) = 0.$$

$$\Rightarrow dS_t = \sigma dW_t.$$

Cash Flow Rate:

$$\hat{S}_t v_t = (S_t - kv_t)v_t = S_t v_t - kv_t^2.$$

Objective Function:

maximize expected proceeds,

$$J(v) = \mathbb{E} \int_0^T (S_u v_u - kv_u^2) du.$$

Value Function:

$$H(t, s, q) = \sup_{v \in \mathcal{A}} \mathbb{E}_{t, s, q} \left(\int_t^T (S_u v_u - kv_u^2) du \right),$$

with HJB

$$\frac{\partial H}{\partial t} + \frac{\sigma^2}{2} \frac{\partial^2 H}{\partial s^2} + \max_v \left\{ -v \frac{\partial H}{\partial q} + sv - kv^2 \right\} = 0,$$

$$\text{with } H|_T = -\infty \cdot \mathbb{1}_{q > 0}$$

Ansatz: $H(t, s, q) = qs + h(t, q)$,

(mark-to-market qs + relative value h).

HJB simplifies,

$$\frac{\partial h}{\partial t} + \max_v \left\{ -kv^2 - v \frac{\partial h}{\partial q} \right\} = 0$$

where

$$\frac{\partial^2 H}{\partial s^2} = \frac{\partial^2}{\partial s^2} (qs + h) = 0$$

$$-v \frac{\partial H}{\partial q} = -v \frac{\partial}{\partial q} (qs + h) = -vs - v \frac{\partial h}{\partial q}$$

yellow terms cancel.

FOC,

$$-2kv - \frac{\partial h}{\partial q} = 0$$

$$\text{or } \boxed{v^* = -\frac{1}{2k} \frac{\partial h}{\partial q}}$$

Insert v^* into HJB,

$$\frac{\partial h}{\partial t} + \frac{1}{4k} \left(\frac{\partial h}{\partial q} \right)^2 = 0.$$

Ansatz: $h(t, q) = -h_2(t)q^2$

$$\Rightarrow -\frac{d}{dt} h_2(t) + \frac{h_2^2(t)}{k} = 0.$$

We need $h_2(T) = \infty$,

which we can get after solving.

$$\frac{\frac{d}{dt} h_2(t)}{h_2^2(t)} = \frac{1}{k}$$

$$\Rightarrow \frac{-1}{h_2(t)} = \frac{t}{k} + C$$

$$\Rightarrow h_2(t) = \frac{k}{-t + C'}$$

is general solution.

Using $h_2(T) = \infty$,

choose $C' = T$,

$$\Rightarrow \boxed{h_2(t) = \frac{k}{T-t}}$$

Insert solution into eqn for v^* ,

$$v^* = -\frac{1}{2k} \frac{\partial h}{\partial q} = \frac{\frac{\partial}{\partial q} (-h_2(t)q^2)}{-2k}$$

$$= \frac{q h_2(t)}{k}$$

$$= \frac{q}{T-t}.$$

Finally, integrate the trajectory,

$$\frac{d}{dt} Q_t^* = -v_t^* = -\frac{Q_t^*}{T-t}$$

$$\Leftrightarrow \frac{\frac{d}{dt} Q_t^*}{Q_t^*} = -\frac{1}{T-t}$$

$$\Leftrightarrow \int_0^t \frac{\frac{d}{ds} Q_s^*}{Q_s^*} = -\int_0^t \frac{1}{T-s}$$

$$\Leftrightarrow \log Q_t^* - \log Q_0$$

$$= \log(T-t) - \log(T)$$

$$\Leftrightarrow \log Q_t^* = \log \mathcal{N} + \log \left(\frac{T-t}{T} \right)$$

$$\Leftrightarrow \boxed{Q_t^* = \mathcal{N} \left(1 - \frac{t}{T} \right)}$$

Achieves a Time Weighted

Average Price (TWAP).

Easy to see,

$$v_t^* = \frac{1}{T} \mathcal{N}.$$

• Expected Execution cost

$$EC^* = \mathbb{E} \int_0^T k(v_t^*)^2 dt = \frac{k\mathcal{N}^2}{T}$$

• For risk-neutral agent,

TWAP is optimal ($v_t^* = \mathcal{N}/T$)

$$\frac{1}{T} \int_0^T v_t^2 dt \geq \left(\frac{1}{T} \int_0^T v_t dt \right)^2 = \frac{\mathcal{N}^2}{T^2}$$

because $\int_0^T v_t dt = \mathcal{N}$

(recall $Q_t = Q_0 - \int_0^t v_s ds$).

• Timing is not important

because $\mathbb{E} dS_t = 0$.